

Accessible Slides in LaTeX

https://github.com/rhstanton/accessible_LaTeX

Version 1.4

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Introduction

- On January 8, 2026, we were notified by campus that, beginning in April 2026,
“The updated requirements of the ADA require that digital course materials provided to students, even materials inside password-protected course sites like bCourses, will need to comply with accessibility standards (Web Content Accessibility Guidelines ([WCAG](#)) 2.1 Level AA).”
- Many of us use $\text{\LaTeX}$ to create teaching materials—both slides and documents.
- Standard $\text{\LaTeX}$ (including Beamer) does not automatically generate accessible PDFs.
- **The good news:** The $\text{\LaTeX}$ Project team has developed solutions!

What is this project?

- An experiment exploring accessible $\LaTeX$ documents using the $\LaTeX$ Tagging Project
 - **Two templates:** Articles (`article` class) and slides (`ltx-talk`)
 - **Shows:** What's common to both, what's different
 - **Contains:** Working examples with math, text, graphics, and tables
 - **Scores:** Perfect 100% from the bCourses accessibility checker (Ally)
- **How to use it:**
 1. **Learn the common requirements:** Apply to any $\LaTeX$ document
 2. **See class-specific needs:** Understand slides vs. articles
 3. **Copy and adapt:** Use as templates for your documents
 4. **Study the code:** Heavily commented for learning
- Available at: https://github.com/rhstanton/accessible_LaTeX

Converting to accessible LaTeX: Overview

- **Your existing $\LaTeX$ skills transfer** - you're just adding accessibility features
- **The changes are minimal** and follow consistent patterns
- **Common requirements** (both slides & articles):
 - Add document metadata and enable PDF tagging
 - Tag images with alt text; tag table headers
 - Use accessible colors; compile with LuaLaTeX
- **Key difference:**
 - Articles: **Keep your existing class!** Just add accessibility features
 - Slides: Switch to `ltx-talk` (similar syntax, but recreate styling)
- Both require LaTeX kernel 2025-11-01 or later (see next slides for details)
- See “Getting started” slides at end for detailed steps
- [LaTeX Tagging Project documentation](#)

TeX version requirements (both slides and articles)

- **Minimum:** TeX Live 2023 or later with all packages updated
- **Critical:** Must have LaTeX kernel 2025-11-01 or later
 - Update via TeX Live Manager (Windows) or TeX Live Utility (Mac)
 - Or use Overleaf Labs (see next slide)
- **Will NOT work:** TeX Live 2022 or earlier

You **CAN** use Overleaf (both slides and articles)

- `ltx-talk` requires a very recent TeX Live version
- This is available through Overleaf's [Labs program](#) (not standard Overleaf)
- **Using Overleaf (2 steps)**
 1. **Join Overleaf Labs:**
 - Visit <https://www.overleaf.com/labs/participate>
 - Opt in and enable "Rolling TeX Live releases"
 2. **Configure project:**
 - Set TeX Live version to "Rolling TeXLive (labs)" (bottom of list)
 - Set Compiler to **LuaLaTeX**
- **Resources:** <https://docs.overleaf.com/writing-and-editing/creating-accessible-pdfs>

Common requirements (1/3): Document Metadata

- **REQUIRED** for both slides AND articles
- Every accessible document **must** begin with `\DocumentMetadata`
- Goes *before* `\documentclass`

```
\DocumentMetadata{
  pdfstandard=a-2u,      % PDF/A-2u format
  pdfstandard=ua-1,      % Add PDF/UA-1 conformance target
  pdfversion=1.7,        % Explicit PDF version
  lang=en-US,            % Language for screen readers
  tagging=on,            % Enable PDF tagging
  tagging-setup={
    math/alt/use,         % Keep math accessibility enabled
    math/mathml/AF=false, % Do not embed MathML as PDF Associated Files
    math/tex/AF=false,    % Do not embed TeX source as Associated Files
    math/mathml/sources=  % Clear MathML source attachment list
  }
}
```

- Configures accessibility, math tagging, and validator-compatibility behavior in one place.

Common requirements (2/3): Images, Tables and Colors

- **REQUIRED for both slides AND articles**
- **Images:** All images must include alt text descriptions
 - Allows screen readers to describe visual content
 - Even decorative images need marking
 - See dedicated “Figures” slide for examples
- **Tables:** Must specify which rows are headers
 - Helps screen readers navigate table structure
 - Essential for data accessibility
 - See dedicated “Tables” slide for examples
- **Accessible colors:** WCAG 2.1 requires 4.5:1 contrast ratio

```
\colorlet{AccessibleRed}{red!80!black}  
\colorlet{AccessibleGreen}{green!40!black}  
% Standard blue is fine as-is
```

Common requirements (3/3): LuaLaTeX

- **REQUIRED for both slides AND articles**
- You **must** compile with LuaLaTeX (not pdfLaTeX or XeLaTeX)
- **Why LuaLaTeX?**
 1. **Automatic MathML:** Makes math accessible without manual work
 2. **Full tagging support:** Complete PDF accessibility features
 3. **Modern fonts:** Handles Unicode and OpenType fonts
- **How to switch:**
 - Command line: `lualatex filename.tex`
 - Most editors: Select “LuaLaTeX” from compiler menu

The basics

- Slides are put inside a `frame` environment, just like in Beamer.
- So **existing source files don't need a lot of editing.**
- Here's some gratuitous *math* for the accessibility checker.

```
\begin{frame}{The basics}
\begin{itemize}
\item Slides are put inside a \texttt{frame} environment, just like in Beamer.
\item So \textbf{\color{blue}existing source files don't need a lot of editing.}
\item Here's some gratuitous  $\mathit{math}$  for the accessibility checker.
\end{itemize}
\end{frame}
```

- **Note:** I set the fonts in this file so that numbers, percent signs, and dollar signs look the same in both math and text mode (my pet Beamer peeve...)
 - **Text:** \$1234567890%.
 - **Math:** \$1234567890%.

Figures

- Including figures is the same as in Beamer (e.g., using `\includegraphics`), but you need to provide a **text description**.

```
\includegraphics[height=.4\textheight,alt={A capybara}]{capybara.jpg}
```



Tables

- Including tables is the same as in Beamer (e.g., using the `tabular` environment).
- Also need to specify **header rows**. Use `{1}` for 1 header row, `{1,2}` for 2 rows, etc.
- **Note:** This example may still trigger PAC's Table header cell has no associated subcells warning even when the source tagging is correct.

```
\tagpdfsetup{table/header-rows={1,2,3}}  
\begin{tabular}{cccccrrrr}  
\toprule  
← 3 header rows  
\midrule  
← data rows  
\bottomrule  
\end{tabular}
```

Payment date	Caplet expiry date	DF_{pay}	Forward rate	Days to expiry	Days in accrual period	T_{expiry}	Δ	Caplet
2004/12/01	—	0.99550	0.01790	0	91	0.00000	0.25278	—
2005/03/01	2004/11/29	0.99008	0.02188	89	90	0.24384	0.25000	1,178.77
2005/06/01	2005/02/25	0.98401	0.02413	177	92	0.48493	0.25556	4,844.73
2005/09/01	2005/05/27	0.97733	0.02675	268	92	0.73425	0.25556	10,016.71

Common pitfalls

- Forgetting alt text for images
 - Every `\includegraphics` needs an `alt={...}` parameter
 - Even decorative images need alt text (use `alt={decorative}`)
- Not specifying table header rows
 - Add `\tagpdfsetup{table/header-rows={...}}` before each table
 - Use `{1}` for 1 header row, `{1,2}` for 2 header rows, etc.
- Insufficient color contrast
 - WCAG 2.1 requires 4.5:1 contrast ratio for normal text
 - Avoid light colors: yellow, cyan fail contrast requirements
 - Darken red and green: use `red!80!black`, `green!40!black`
 - Standard blue is fine and meets WCAG requirements
 - Test with a contrast checker: <https://webaim.org/resources/contrastchecker/>
- Using the wrong compiler
 - Make sure your editor is set to use `LuaLaTeX`, not `pdfLaTeX`
- Old TeX distribution
 - TeX Live 2022 or earlier won't work
 - Update packages using TeX Live Manager (Windows) or TeX Live Utility (Mac)

Getting started: Common steps

1. **Setup environment:** Use Overleaf Labs OR install/update TeX Live locally
 - Overleaf: See earlier slides for Labs setup
 - Local: Update via TeX Live Manager/Utility
2. **Get templates:** Download from https://github.com/rhstanton/accessible_LaTeX
3. **Add accessibility features:**
 - Add alt text to images
 - Add table/header-rows to tables
4. **Set compiler to LuaLaTeX**
5. **Compile:**
 - Compile with LuaLaTeX
 - PDF file now scores 100% on Ally

Getting started: What's different by document type

Articles (using article, report, book, etc.):

- Add `\DocumentMetadata` before `\documentclass`
- Include the tagging-setup block in `\DocumentMetadata` (as shown in this template)
- Switch to `fontspec` and `unicode-math` packages
- Keep your existing `documentclass`!

Slides (migrating from Beamer):

- Copy preamble from `accessible_slides.tex`
- Change `\documentclass{beamer}` to `\documentclass[frame-title-arg]{ltx-talk}`
- Remove Beamer themes/templates/colors
- Recreate styling using standard LaTeX/xcolor
- One-time preamble work—then reuse for all future talks!
- Questions? richard.stanton@berkeley.edu

Validation workflow: use multiple checkers

- No single checker catches everything.
- Accessibility usability checks and strict PDF conformance checks are not identical.
- A practical workflow is to run more than one checker and compare findings.
- **Checkers used so far**
 - Ally (built into bCourses)
 - veraPDF (open-source PDF conformance validator, <https://verapdf.org/>)
 - PAC (PDF Accessibility Checker, <https://pac.pdf-accessibility.org/>)
 - Adobe Acrobat.

Optional validator cleanup step: strip_af.py

- The metadata settings alone are enough to generate a 100% score on Ally.
- Stricter checkers (e.g., veraPDF) may still complain about /AF or embedded-file structures.
- To remove these and stop the complaints, run strip_af.py:

```
python strip_af.py build/accessible_slides.pdf
```

- Prerequisite: install the Python package pikepdf first: `python -m pip install pikepdf`.
- Then validate build/accessible_slides.pdf with veraPDF.

Validator-specific notes

- **Acrobat: “Lb1 and LBody – Failed” (validator bug)**
 - Appears around figure captions, section numbers, table captions
 - This is an Acrobat bug, not a source error – safe to ignore
 - The <Lb1> tag is correctly used for numbering
 - Acrobat’s checker incorrectly expects it only in lists
 - Reference: <https://tex.stackexchange.com/a/709655/2388>
- **PAC table warning (known behavior)**
 - PAC can report `Table header cell has no associated subcells` even when `table/header-rows` is correct
 - This behavior is documented by the LaTeX Tagging Project: <https://github.com/latex3/tagging-project/issues/1056>
- **When this warning appears**
 - First confirm source markup is correct (especially `table/header-rows`)
 - Keep the source tagging and document the validator warning
 - Report minimal reproductions to tool vendors when possible
- **Bottom line**
 - Use multiple validators (Ally, veraPDF, PAC, Acrobat) and compare findings
 - Treat one-tool warnings as signals to investigate, not proof of errors